

# CS253 - Data Structures and Algorithms for DS/AI

## Spring 2025

PSO 9, Week 10

### Question 1

**(2-3 tree)**

- (1) How many 2-3 trees exist storing the keys  $\{1, 2, 3, 4, 5\}$ ? Explain your answer.
- (2) Bruno says the largest key in a 2-3 tree can sometimes be found in an interior node. Is Bruno right? Explain your answer.

**Question 2****(Graphs)**

1. True or False:

- (1) There exists a simple, undirected graph with 5 nodes, each of degree 3.
- (2) There exists a simple, undirected graph  $G$  with  $n$  vertices, whose vertex degrees are  $0, 1, 2, \dots, n-1$ .  
(assume  $n > 3$ )

2. A tree is the most widely used special type of graph, in a sense that it is the minimal connected graph. Prove the following important lemma:

*Let  $G$  be an undirected graph, any two of the following properties imply the third property, and that  $G$  is a tree.*

- (1)  $G$  is connected;
- (2)  $G$  is acyclic;
- (3)  $G$  satisfies  $|E| = |V| - 1$ .